EXP NO:16

Experiment to understand EDA-Quantitative and

AIM:

Understand the importance of EDA-Quantitative and Qualitative analysis.

PROGRAM:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

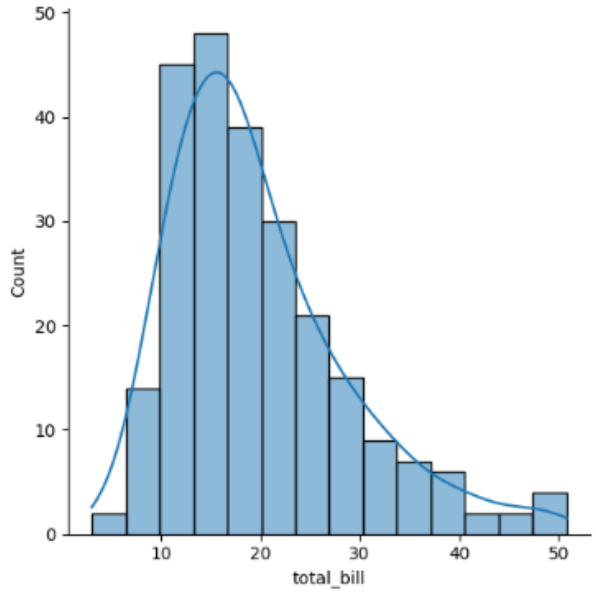
%matplotlib inline

tips = sns.load_dataset('tips')
tips.head()
sns.displot(tips.total_bill,kde=True)
sns.displot(tips.total_bill,kde=False)
sns.jointplot(x=tips.tip,y=tips.total_bill)
sns.jointplot(x=tips.tip,y=tips.total_bill)
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
sns.pairplot(tips)
tips.time.value_counts()
sns.pairplot(tips,hue='day')
sns.heatmap(tips.corr(numeric_only=True),annot=True)
sns.boxplot(tips.total_bill)
sns.boxplot(tips.tip)
sns.countplot(tips.day)
sns.countplot(tips.sex)
tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')
sns.countplot(tips[tips.time=='Dinner']['day'])
```

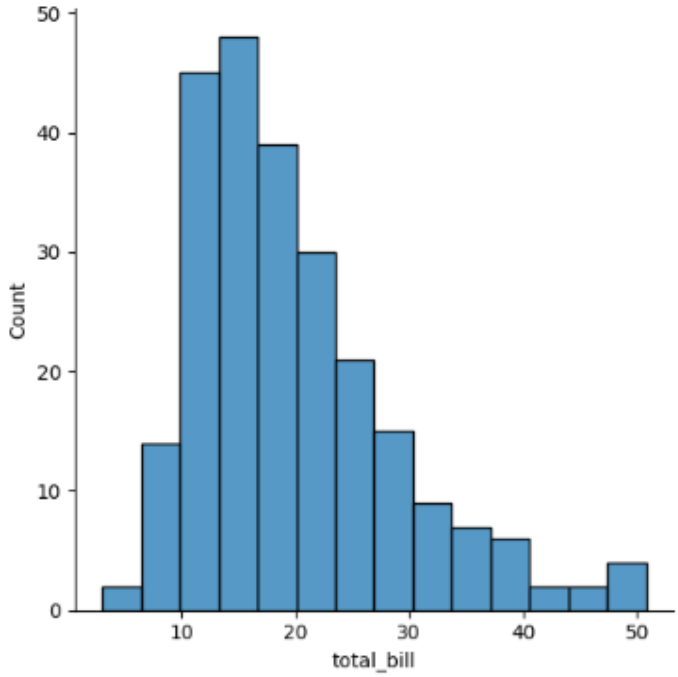
OUTPUT:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

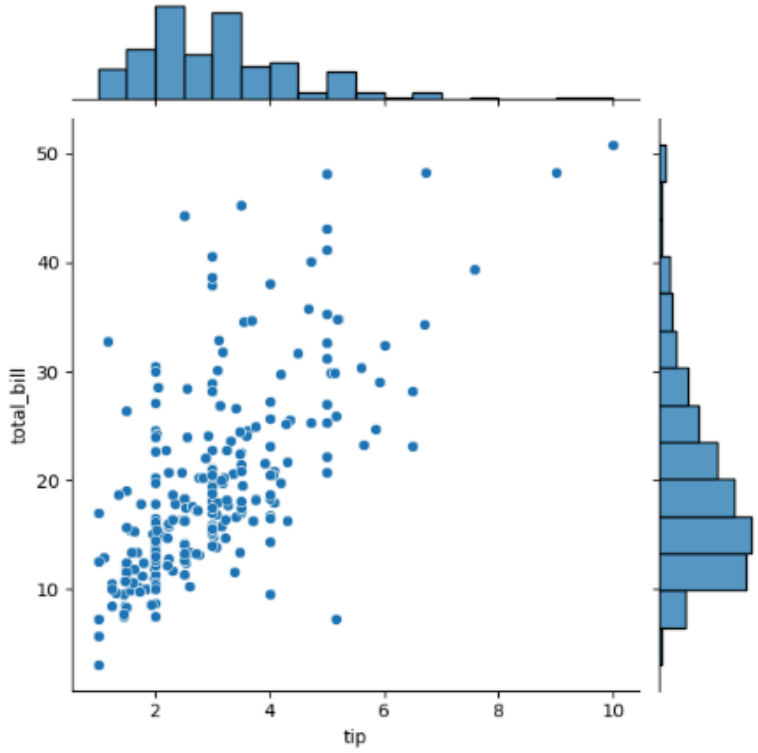
```
<seaborn.axisgrid.FacetGrid at 0x7ca737b95940>
```



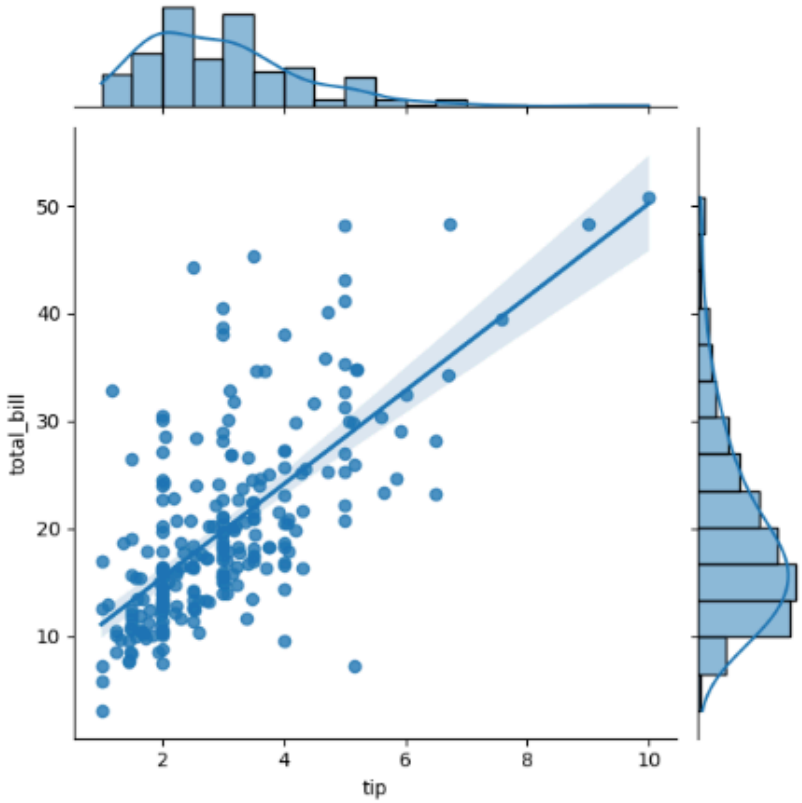

```
<seaborn.axisgrid.FacetGrid at 0x7ca7379cc690>
```



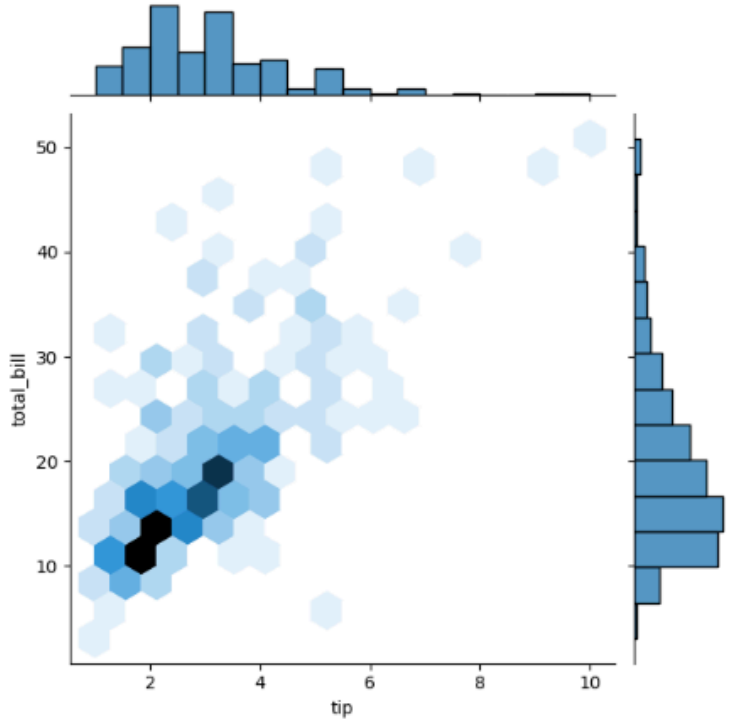
```
<seaborn.axisgrid.JointGrid at 0x7ca737a55400>
```



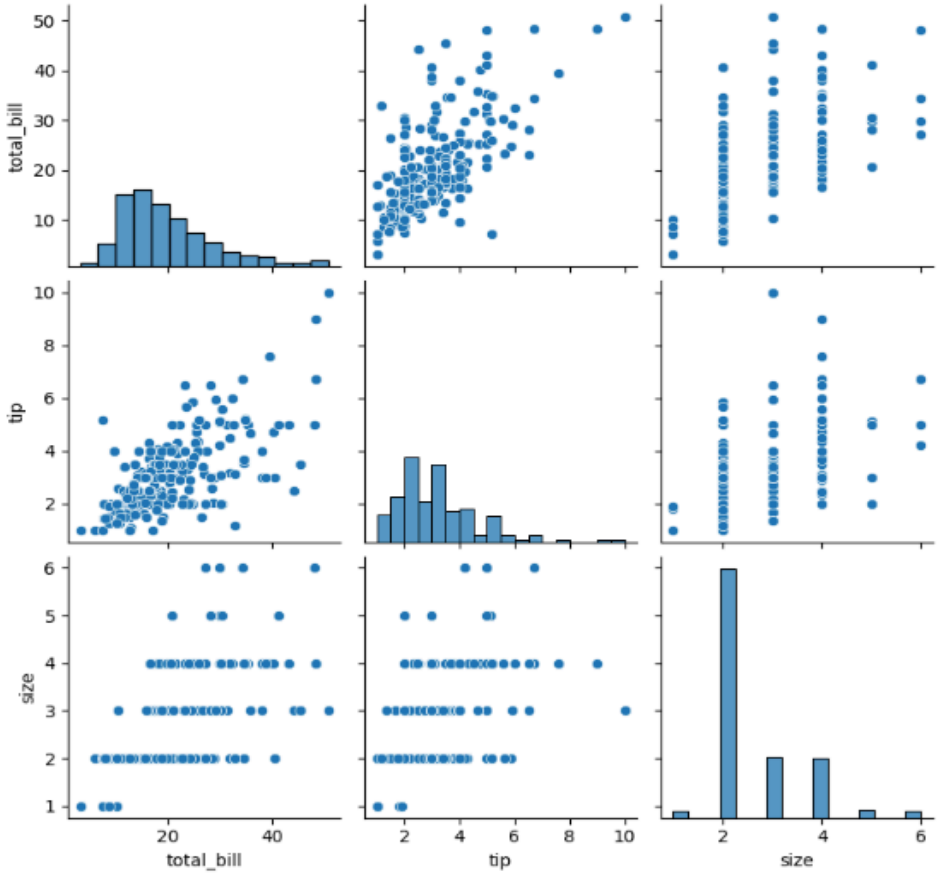

```
<seaborn.axisgrid.JointGrid at 0x7ca7335456d0>
```



```
<seaborn.axisgrid.JointGrid at 0x7ca73346b110>
```



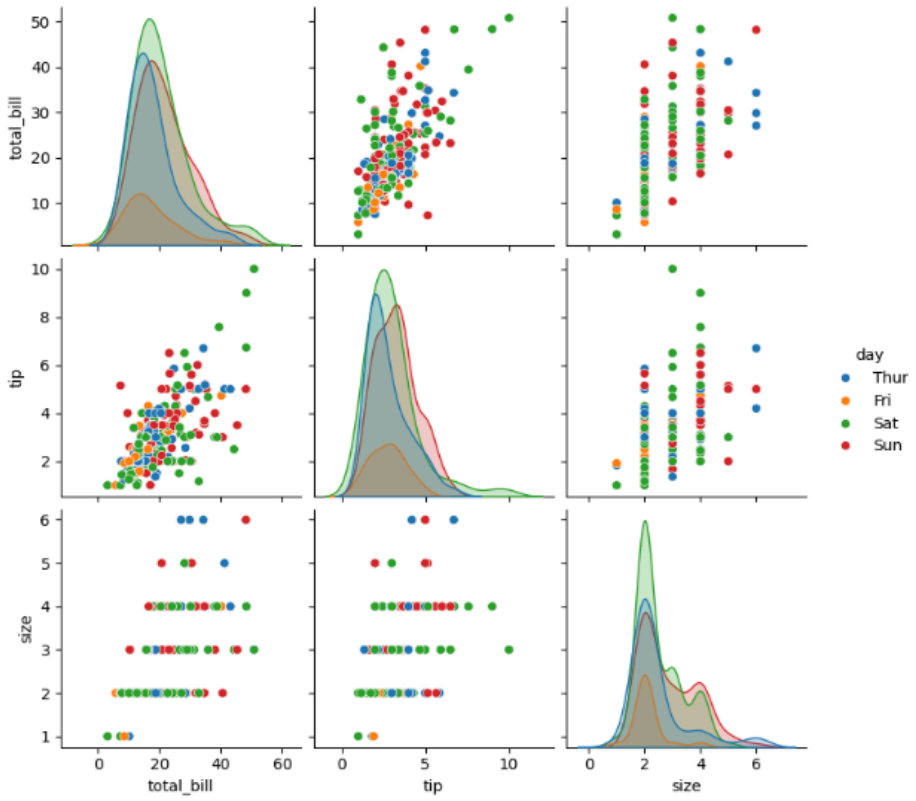

```
<seaborn.axisgrid.PairGrid at 0x7ca737a56660>
```



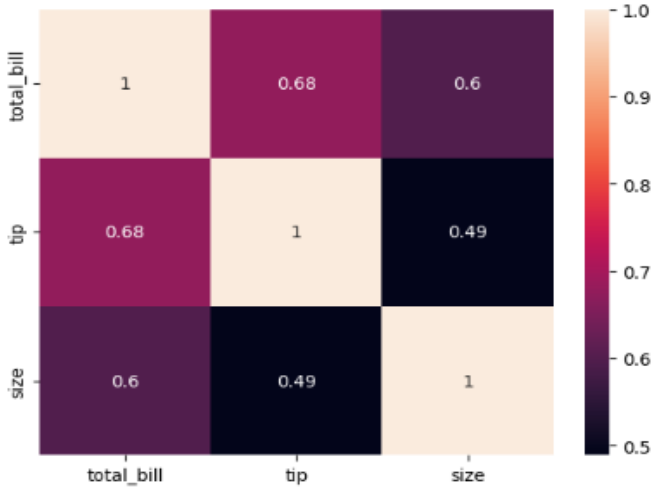
time	count
Dinner	176
Lunch	68

dtype: int64

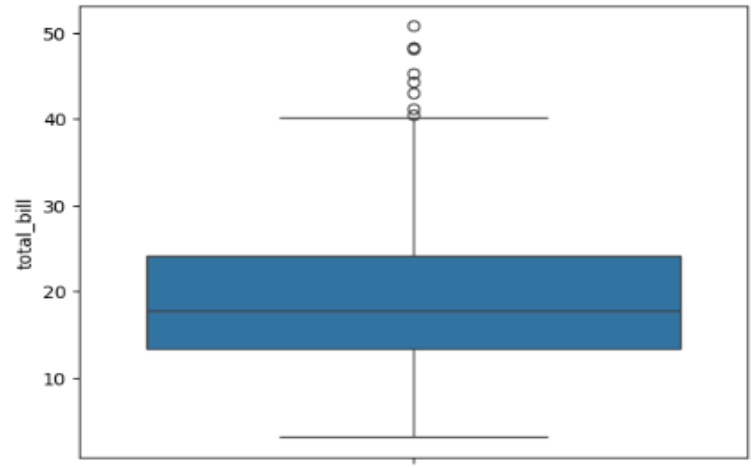
```
<seaborn.axisgrid.PairGrid at 0x7ca73143e850>
```



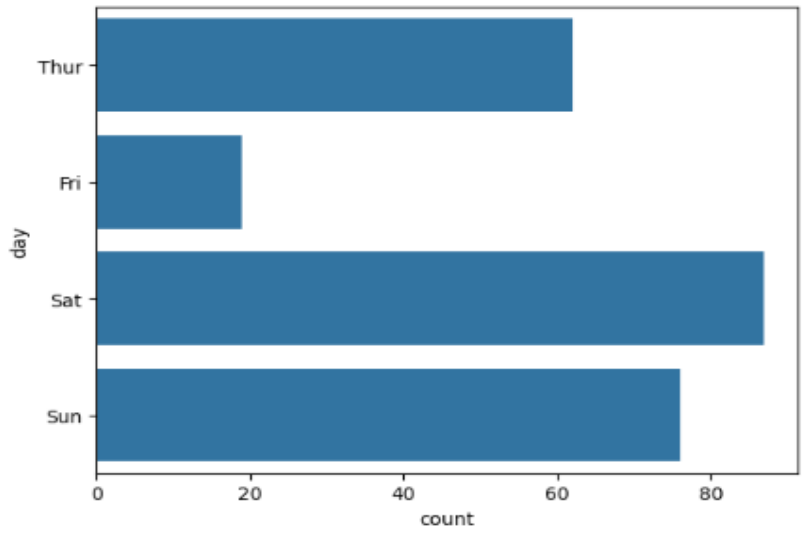

```
<Axes: >
```



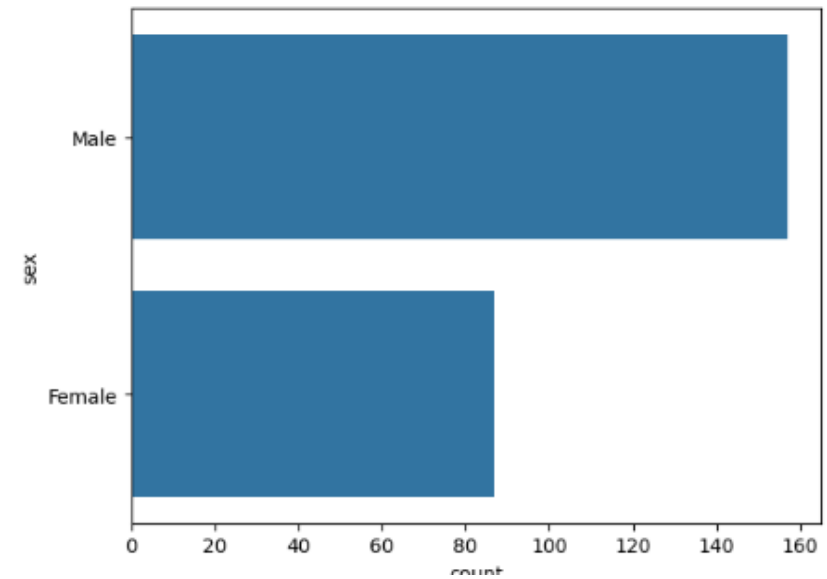
```
<Axes: ylabel='total_bill'>
```



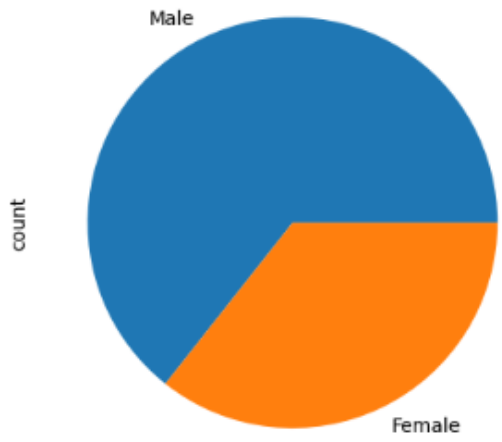

```
<Axes: xlabel='count', ylabel='day'>
```



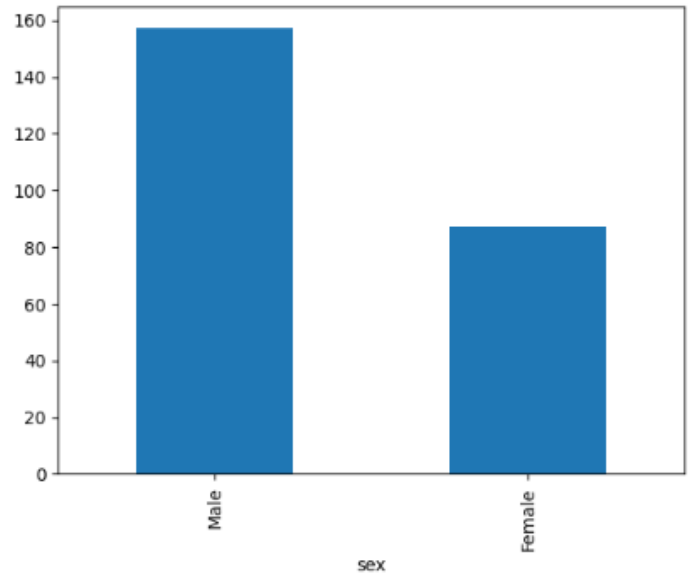

```
<Axes: xlabel='count', ylabel='sex'>
```



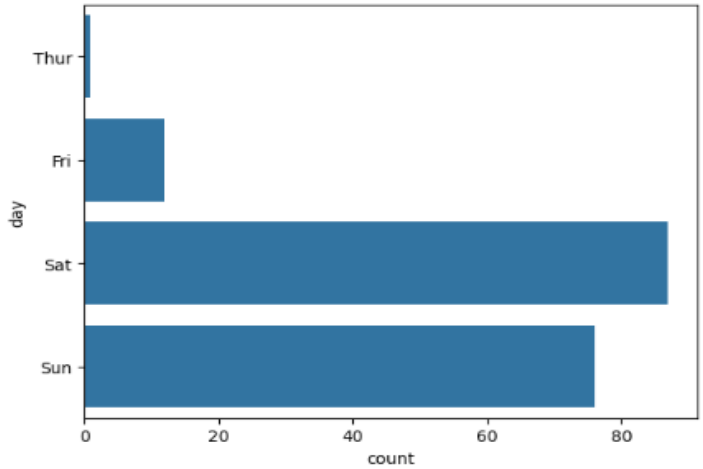
```
<Axes: ylabel='count'>
```




```
<Axes: xlabel='sex'>
```




```
<Axes: xlabel='count', ylabel='day'>
```



RESULT:

The experiment has been carried out successfully